

MACH24 Supplementary Material

Supplementary methods tables and figures

Supplementary methods

The sequencing denominator contained 24,140 unique participants. Eligibility required presence in this denominator and age at blood collection of at least 18 completed years. Participants were linked to clinical records by normalized medical record number, with sample and laboratory identifiers retained for cross-checking. Age was calculated using full-year birth dates from the demographic records and the index blood-collection date. Fourteen participants were younger than 18 years, leaving 24,126 adults (Figure 1). No additional exclusion based on hematological diagnoses, missing diagnosis dates or billing records was applied in this release.

Eligibility in this release was restricted by sequencing-denominator membership and adult age, without hematological malignancy exclusions or clinical adjudication of all CH categories. Conventional CHIP is therefore an operational VAF-based category rather than confirmation that every carrier meets all clinical diagnostic criteria for CHIP. Studies requiring absence of a pre-existing hematological malignancy need additional disease-specific eligibility assessment.

Variant filtering and sequencing. Primary retention required missing sensitivity_alt_reads, recurrence_flag and pon flags, and VAF between 0.003 and 1 inclusive. This retained 15,274 records from 9,904 participants: 10,840 at $VAF < 2\%$ and 4,434 at $VAF \geq 2\%$. Low-VAF CH only required a retained variant with $0.3\% \leq VAF < 2\%$ and none at $VAF \geq 2\%$; conventional CHIP required at least one retained variant at $VAF \geq 2\%$. Participants with no retained variants were classified as no detected CH. Sequencing-denominator membership, not variant-row presence, defined sequencing eligibility. The retained export contained no duplicate participant-variant identifiers and no participant with retained records from multiple samples.

Descriptive and regression analyses. Participant proportions use two-sided 95% Wilson intervals where displayed; tumor-site plots report observed proportions without intervals. Separate logistic models compare low-VAF CH only and conventional CHIP with no detected CH, adjusting for age per 10 years, sex, recorded race and smoking. These associations are descriptive and not causal.

Follow-up. Follow-up was measured from the index blood draw to the verified date of death or, for participants without a recorded death, the last recorded contact. A valid observed endpoint was available for 24,125 of 24,126 participants. Median observed follow-up was 5.71 years (range, 0.05–18.54 years), with 8,280 verified deaths after blood collection. Dates were evaluated against a data-review cutoff of 16 September 2026. The participant without a valid endpoint was omitted from survival estimation; no contact date was imputed.

Supplementary Tables S1 and S2 (capture-panel summaries and coordinates) are supplied in the accompanying Excel workbook. Their assay-design data are unchanged by cohort eligibility.

Supplementary tables

Supplementary Table S3. Cohort-level genomic and sequencing characteristics. VAF, variant allele fraction; CH, clonal haematopoiesis; CHIP, clonal haematopoiesis of indeterminate potential.

Characteristic	Summary type	Value
Analytic cohort	Participants	24,126
With selected unfiltered sequencing records	Participants	24,117 (99.96%)
Any retained CH variant	Participants	9,904 (41.1%)
Low-VAF CH only	Participants	6,203 (25.7%)
Conventional CHIP	Participants	3,701 (15.3%)
Both low and high VAF variants	Participants	1,616 (6.7%)
No detected CH	Participants	14,222 (58.9%)
No-CH participants with selected unfiltered records	Participants	14,213
No-CH participants without selected unfiltered records	Participants	9
Unfiltered records per no-CH participant with records	Median [IQR]	38.0 [34.0, 42.0]
Participant median depth in no-CH group with records	Median [IQR]	2,684.0 [2,203.0, 3,072.0]×
Low VAF <2% variants	Records	10,840
Low VAF <2% depth	Median [IQR]	2,972.5 [2,467.0, 3,516.0]×
Low VAF <2% alt	Median [IQR]	21.0 [16.0, 31.0]
Low VAF <2% vaf	Median [IQR]	0.78 [0.53, 1.17]%
CHIP VAF ≥2% variants	Records	4,434
CHIP VAF ≥2% depth	Median [IQR]	2,664.5 [2,083.2, 3,201.8]×
CHIP VAF ≥2% alt	Median [IQR]	115.5 [73.0, 235.0]
CHIP VAF ≥2% vaf	Median [IQR]	4.39 [2.82, 8.92]%

Supplementary Table S4. Complete gene-level mutation and carrier landscape. All genes with retained variant records are shown. Variant counts are retained records and are not necessarily counts of unique genomic loci.

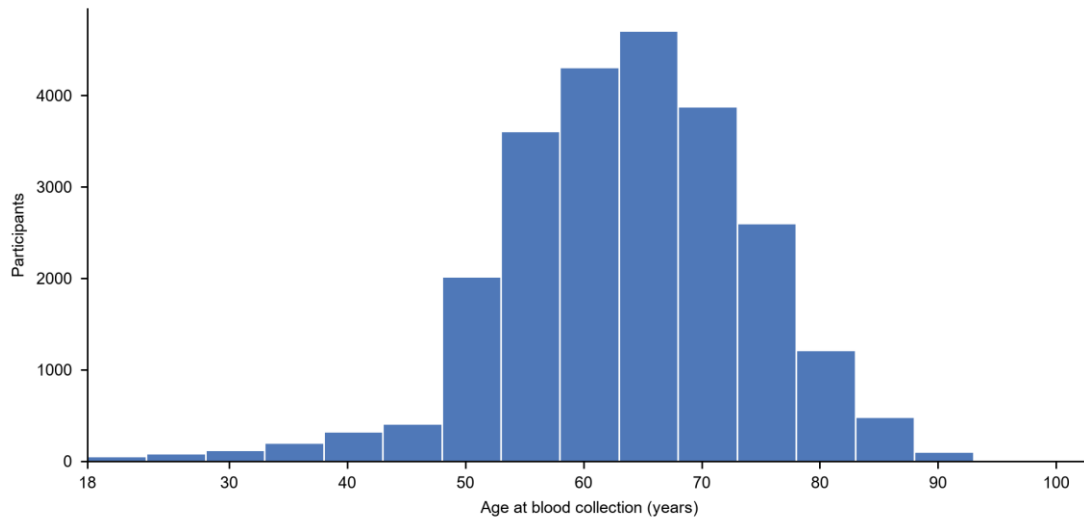
Gene	Low-VAF variants	VAF $\geq 2\%$ variants	Low-VAF carriers	VAF $\geq 2\%$ carriers	Low-VAF fraction
DNMT3A	4628	1947	3904	1831	70.4%
TET2	2333	969	2065	873	70.7%
PPM1D	1292	464	1163	438	73.6%
CHEK2	687	176	650	173	79.6%
TP53	620	189	590	184	76.6%
ASXL1	495	252	479	249	66.3%
CBL	178	33	176	33	84.4%
SF3B1	102	97	100	96	51.3%
GNAS	98	37	98	37	72.6%
GNB1	85	56	85	56	60.3%
JAK2	66	79	66	79	45.5%
BRCC3	60	30	60	29	66.7%
KRAS	51	12	51	12	81.0%
SRSF2	44	54	44	54	44.9%
IDH2	41	19	41	19	68.3%
NRAS	25	3	25	3	89.3%
IDH1	18	5	18	5	78.3%
ASXL2	8	4	8	4	66.7%
MPL	7	2	7	2	77.8%
KIT	1	0	1	0	100.0%
SETBP1	1	6	1	6	14.3%

Supplementary Table S5. Cohort assembly. Counts are unique participants; age is measured in completed years at collection. No diagnosis or billing exclusion was applied.

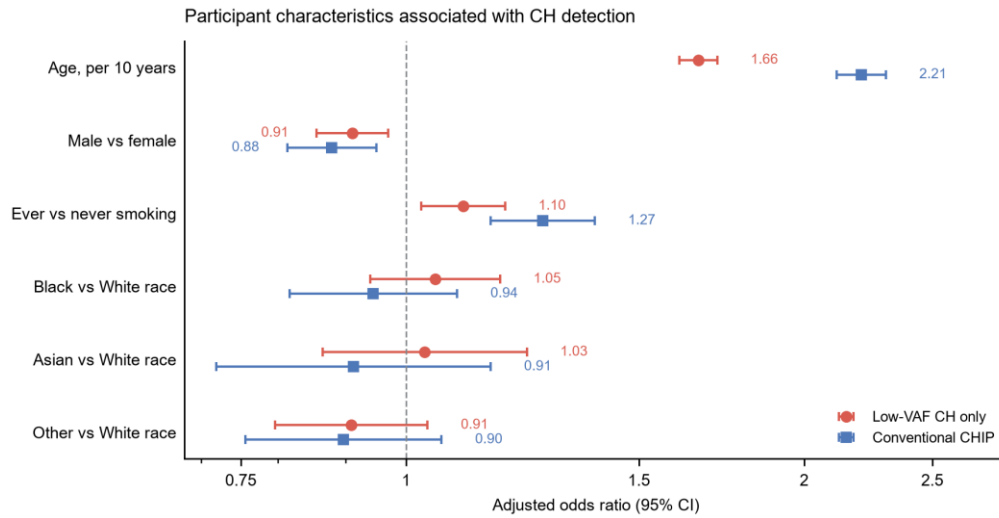
Eligibility stage	Remaining participants	Excluded at step
Sequencing denominator	24140	0
Age at collection ≥ 18 years	24126	14

Age distribution at blood collection

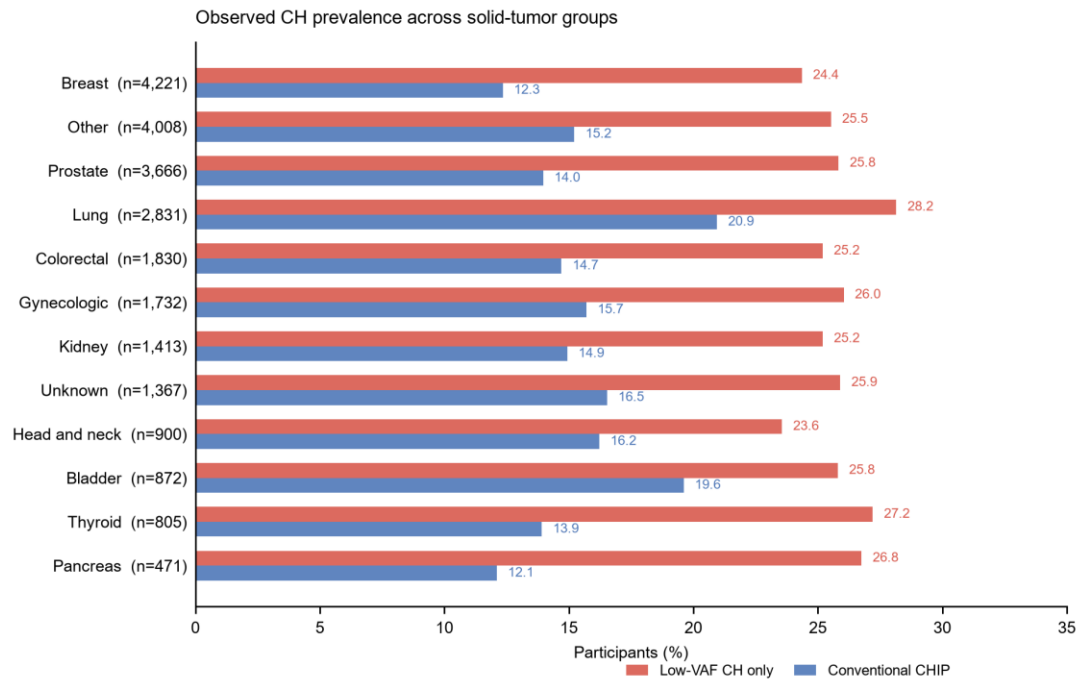
n = 24,126 | Median 64.1 years | IQR 57.0–70.6



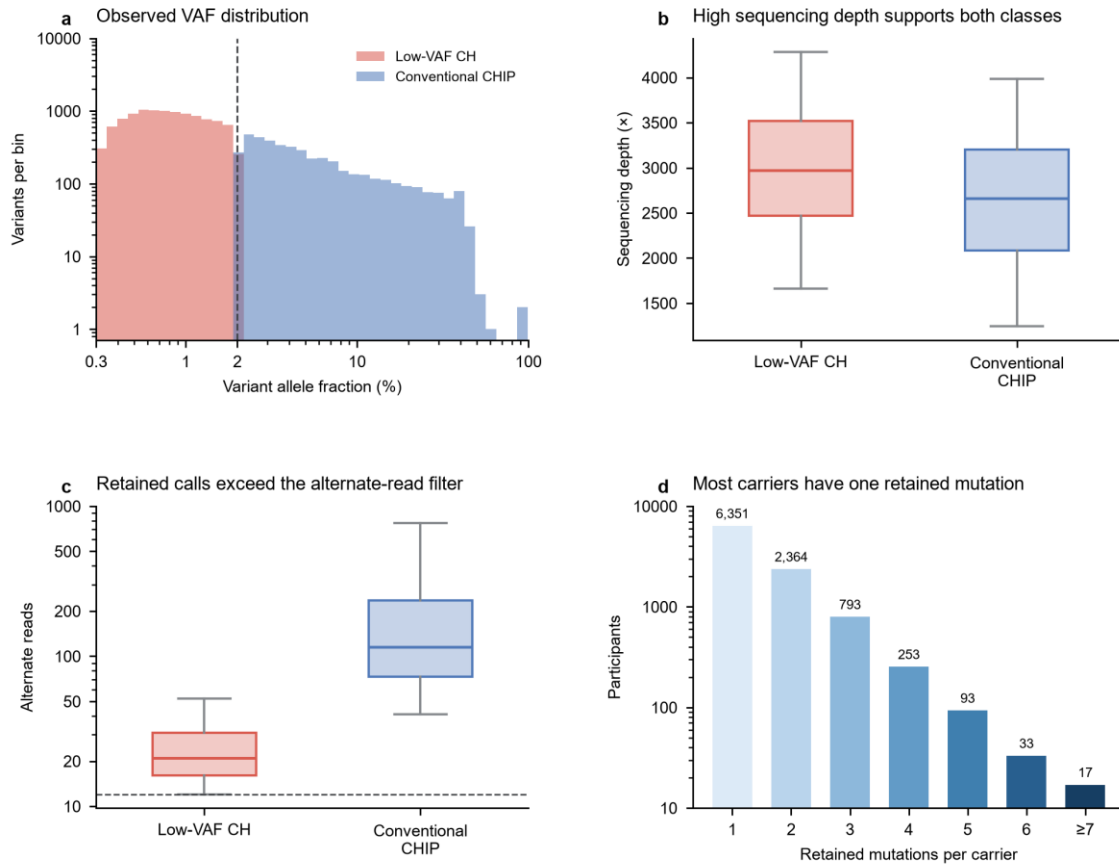
Supplementary Figure S1. Age distribution at blood collection. All 24,126 adults are included, with no missing ages. Bars show counts in five-year intervals beginning at 18 years (18 to <23, 23 to <28, and so forth). Median age was 64.1 years (IQR, 57.0–70.6). Counts are descriptive; no confidence intervals or hypothesis tests are shown.



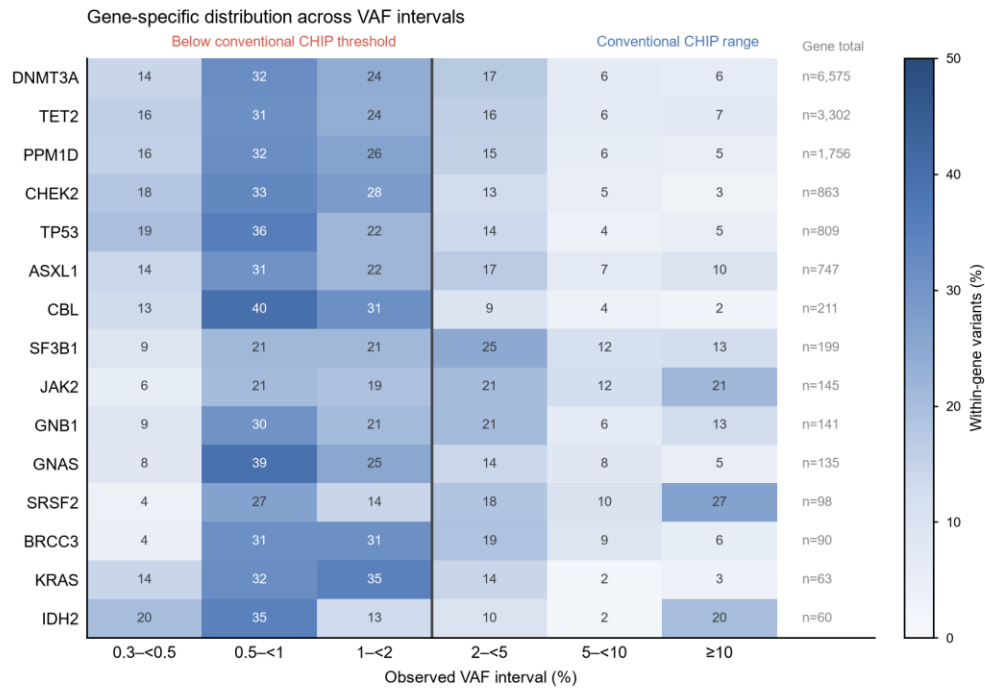
Supplementary Figure S2. Adjusted associations with baseline CH category. Separate logistic models compare low-VAF CH only (6,203 cases) and conventional CHIP (3,701 cases) with no detected CH (14,222 controls), adjusting simultaneously for age, sex, race and smoking status. Points and error bars show odds ratios and two-sided Wald 95% confidence intervals. Age is expressed per 10 years; reference categories are female sex, White race and never smoking. Unknown race and smoking are included as categories. The dashed line marks an odds ratio of 1. These baseline associations do not establish causality.



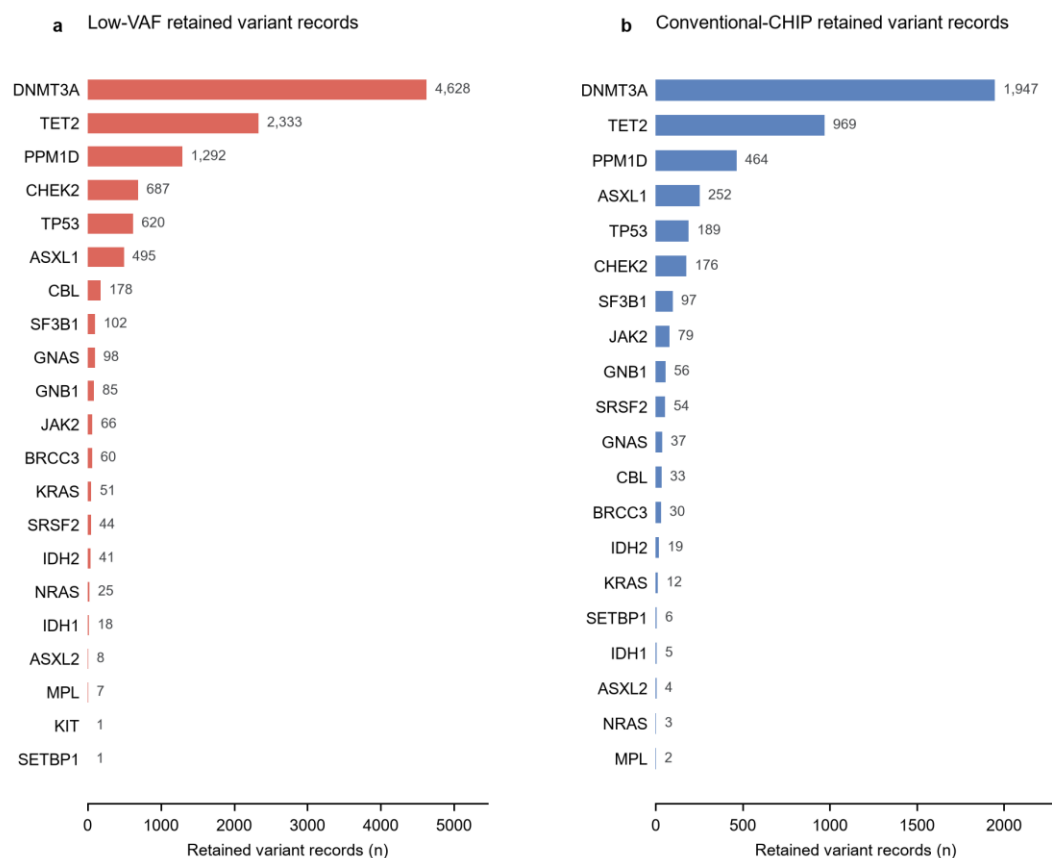
Supplementary Figure S3. Observed CH prevalence across recorded solid-tumor groups. Bars show participant proportions with low-VAF CH only and conventional CHIP, with denominators beside each group. Groups with at least 400 participants are displayed; melanoma (n=10) remains in the aggregate source table. Tumor assignment prioritizes registry primary malignant solid-tumor records, then pre-draw records and proximity to collection. These unadjusted summaries describe the enrolled cohort and are not population-representative estimates. No confidence intervals or hypothesis tests are shown.



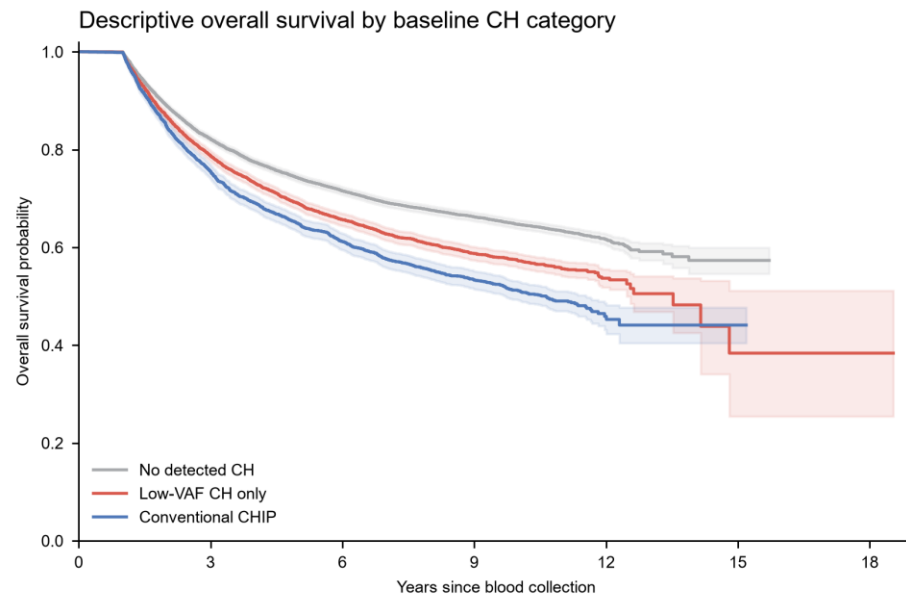
Supplementary Figure S4. Technical support and mutation multiplicity. a, VAF distribution for all 15,274 retained variants (10,840 below 2% and 4,434 at or above 2%); both axes are logarithmic and the dashed line marks 2%. b, Sequencing depth. c, Alternate-read counts, with a dashed reference at 12 reads. Boxes show quartiles, centre lines medians and whiskers the 5th and 95th percentiles. Median depth was 2,972.5× and 2,664.5×, and median alternate-read counts were 21 and 115.5 for the two VAF strata, respectively. d, Retained-variant multiplicity among 9,904 carriers, combining counts of seven or more. All retained records were used without subsampling. Technical support does not replace formal assay validation.



Supplementary Figure S5. Gene-specific distribution of retained variants across observed VAF intervals. The heatmap shows the percentage of each gene's retained variants in six mutually exclusive VAF intervals: 0.3–<0.5%, 0.5–<1%, 1–<2%, 2–<5%, 5–<10%, and ≥10%. The 15 genes with the most retained variants are shown. Percentages use the total number of retained variants in each gene as the denominator. Each row therefore sums to 100%, apart from rounding. Gene-specific totals appear on the right. The vertical line separates intervals below the conventional 2% CHIP threshold from those at VAF ≥2%. Cell values are descriptive percentages; no hypothesis tests were performed. Source data are provided in the accompanying source-data file.



Supplementary Figure S6. Gene rankings by retained-variant count within each VAF stratum. a, Genes with at least one retained variant at VAF<2%. b, Genes with at least one retained variant at VAF≥2%. Numbers beside bars show retained-record counts. The panels cover 10,840 and 4,434 records, respectively, and use different horizontal ranges. Rankings are descriptive; no gene-to-gene hypothesis tests were performed.



No detected CH	14,222	10,879	7,288	5,266	588	18	0
Low-VAF CH only	6,203	4,514	2,855	1,897	183	6	1
Conventional CHIP	3,700	2,552	1,505	977	74	1	0

Supplementary Figure S7. Overall survival by baseline CH category. Kaplan–Meier estimates include 24,125 participants with valid observed follow-up and 8,280 deaths. One participant without a valid endpoint is omitted; no endpoint is imputed. Death is the event and remaining participants are censored at last recorded contact. The data-review cutoff is 16 September 2026. Shading shows pointwise 95% log-log confidence intervals using Greenwood standard errors. Numbers below the plot are participants at risk immediately before each displayed time. Curves end at each group’s last observed follow-up. Comparisons are unadjusted, with no hypothesis test, and do not establish independent or causal effects.